Let A, B, C be sets.

Prove $A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C)$

Let $x \in A \setminus (B \setminus C)$. This means $x \in A$ and $x \notin (B \setminus C)$.

Because $x \notin B \setminus C$, we know $x \in B$ and $x \notin C$ is false.

The negation of "$x \in B$ and $x \notin C$" is "$x \notin B$ or $x \in C$".

We now have $x \in A$, and $(x \notin B$ or $x \in C)$.

Using the distributive law for statements (Problem 13), we then have $x \in A$ and $x \notin B$, or $x \in A$ and $x \in C$.

(Recall $P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R)$)

This means $x \in A \setminus B$ or $x \in A \cap C$, which proves $x \in (A \setminus B) \cup (A \cap C)$. Repeat for all $x \in A \setminus (B \setminus C)$.

we have shown $A \setminus (B \setminus C) \subseteq$ ~~$A \setminus (B \setminus C)$~~ $(A \setminus B) \cup (A \cap C)$.

Now let $x \in (A \setminus B) \cup (A \cap C)$. This means $x \in A \setminus B$ or $x \in A \cap C$.

Suppose first that $x \in A \setminus B$. Then we know $x \in A$ and $x \notin B$.

Because $x \notin B$, we know $x \in B$ and $x \notin C$ is false, so $x \notin B \setminus C$.

This shows $x \in A$ and $x \notin B \setminus C$, proving $x \in A \setminus (B \setminus C)$.

Suppose instead that $x \in A \cap C$. Then $x \in A$ and $x \in C$. Because $x \in C$, we know $x \in B$ and $x \notin C$ is false. Hence we know $x \notin B \setminus C$.

Because $x \in A$ and $x \notin B \setminus C$, we have $x \in A \setminus (B \setminus C)$. Repeat for all $x \in (A \setminus B) \cup (A \cap C)$

we have shown $(A \setminus B) \cup (A \cap C) \subseteq A \setminus (B \setminus C)$.